Robot Localization and Navigation
ROB-GY 6213
Lab 3 - EKF Localization

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***Abstract*— This study implements an Extended Kalman Filter (EKF) using a kinematic bicycle model to mitigate odometry drift by fusing high-frequency encoder data with ArUco-based vision measurements. Experimental validation demonstrates high precision in linear trajectories, yielding mean errors of 0.106 ± 0.113 m (x), 0.102 ± 0.028 m (y), and $3.13 \pm 1.16^\circ$ (θ). While positional error remained below 0.25 m in complex maneuvers, orientation error increased to $86.04 \pm 55.46^\circ$ during high-curvature turns, highlighting model sensitivity to rapid rotational dynamics. The system proved robust to incorrect initializations and sensor occlusion, utilizing covariance expansion to manage uncertainty during periods of partial observability. Ultimately, the EKF provides a stable localization framework by successfully integrating intermittent global corrections with continuous predictive modeling.**

Fig. 1. Robot build after adding 3rd layer and ArUco Tag

I. INTRODUCTION

A fundamental challenge of robot localization is the uncertainty inherent with both the robot's sensors and motors. While previous labs focused on developing a probabilistic motion model based on bicycle kinematics, dead reckoning using just a motor encoder accumulated error based on wheel slip, mechanical play in the system, and numerical integration drift. This can be counteracted with absolute position measurements, and as such, the EKF (Extended Kalman Filter) [1] was used to provide a robust mathematical framework for fusing continuous, non-linear odometry predictions with periodic, noisy global measurements. In this lab, a camera system using ArUco [2] tags was used to provide global (x,y,θ) measurements, and an EKF is implemented to achieve real-time, bounded-error localization.

This report will detail the mathematical methodology and EKF design, including the derivation of the prediction and correction steps. It will then outline the experimental design, including camera calibration and physical testing procedures. From there it will present experimental results, quantifying localization errors, successes, and hardware limitations. Finally, it will conclude with a summary of the quantified performance claims.

II. METHOD

This section describes the mathematical and technical implementation of the Extended Kalman Filter.

A. System Inputs and State Representation

The EKF processes two primary inputs: control signals and sensor measurements. The control vector $\mathbf{u}_t$ is derived from the robot's physical sensors: wheel Encoders which provide the cumulative tick count, and are used to calculate linear distance s, and the commanded steering angle α.The state of the robot at any time t is represented by the vector $\mathbf{x}_t$:

$$\mathbf{x}_t = \begin{bmatrix} x_t \\ y_t \\ \theta_t \end{bmatrix} \tag{1}$$

where x_t and y_t represent the Cartesian coordinates in meters, and θ_t represents the orientation in degrees.

B. Prediction Step

1) Transition Function: The prediction step estimates the state at the current time step based on the previous state $\mathbf{x}_{t-1}$ and the control input $\mathbf{u}_t$. The nonlinear transition function $g(\mathbf{x}_{t-1}, \mathbf{u}_t)$ updates the pose using the distance traveled s and the change in heading $\Delta\theta$. To improve accuracy, a second-order Runge-Kutta integration (half-angle formula) is used:

$$\hat{\mathbf{x}}_t = \begin{bmatrix} x_{t-1} + s \cdot \cos(\theta_{t-1} + \frac{\Delta\theta}{2}) \\ y_{t-1} + s \cdot \sin(\theta_{t-1} + \frac{\Delta\theta}{2}) \\ \theta_{t-1} + \Delta\theta \end{bmatrix} \tag{2}$$

2) Linearization: Because the transition function is nonlinear, the EKF linearizes the system using Jacobians. The state Jacobian G_x represents the partial derivatives of the transition function with respect to the state:

$$G_x = \frac{\partial g}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & -s \cdot \sin(\theta_{t-1} + \frac{\Delta\theta}{2}) \cdot \\ 0 & 1 & s \cdot \cos(\theta_{t-1} + \frac{\Delta\theta}{2}) \cdot \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

The control Jacobian G_u represents the partial derivatives with respect to the control inputs s and $\Delta\theta$:

$$G_u = \frac{\partial g}{\partial \mathbf{u}} = \begin{bmatrix} \cos(\theta_{t-1} + \frac{\Delta\theta}{2}) & -\frac{s}{2} \cdot \sin(\theta_{t-1} + \frac{\Delta\theta}{2}) \cdot \\ \sin(\theta_{t-1} + \frac{\Delta\theta}{2}) & \frac{s}{2} \cdot \cos(\theta_{t-1} + \frac{\Delta\theta}{2}) \cdot \\ 0 & 1 \end{bmatrix} \tag{4}$$

3) Covariance Matrix: The uncertainty of the estimate is propagated forward by projecting the previous covariance Σ_{t-1} through the linearized model and adding the process noise R_t:

$$\bar{\Sigma}_t = G_x \Sigma_{t-1} G_x^T + G_u R_t G_u^T \tag{5}$$

The process noise matrix R_t is defined by the variances of the distance traveled σ_s^2 and the rotational velocity σ_θ^2:

$$R_t = \begin{bmatrix} \sigma_s^2 + \epsilon & 0 \\ 0 & \sigma_\theta^2 + \epsilon \end{bmatrix} \tag{6}$$

where ϵ is a small constant (10^{-8}) to ensure the matrix remains positive-definite and prevents numerical singularity.

C. Correction Step

1) ArucoTag System: The external measurement $\mathbf{z}_t$ is provided by a vision system utilizing ArUco markers. The system identifies a set of fixed environment markers with known global coordinates and a single marker (ID 7) attached to the robot. The pose of the robot in the world frame is calculated through a three-step transformation chain. First, the system detects N fixed markers in the camera image. Using the known world coordinates of these markers' corners and the camera's intrinsic parameters, the Perspective-n-Point (PnP) algorithm is used to determine the transformation from the world to the camera, T_W^C. The camera's pose in the world frame, T_C^W, is then obtained by inverting this matrix:

$$T_C^W = (T_W^C)^{-1} \tag{7}$$

Robot Pose in Camera Frame: Simultaneously, the system detects the corners of the robot marker. A second PnP solver determines the robot's pose relative to the camera lens, represented by the transformation matrix T_R^C.Global Pose Composition: The robot's pose in the global world frame, T_R^W, is computed by composing these two transformations:

$$T_R^W = T_C^W \cdot T_R^C \tag{8}$$

2) Measurement Function: The final measurement vector $\mathbf{z}_t = [x, y, \theta]^T$ is extracted from the translation components and the yaw angle derived from the rotation sub-matrix of T_R^W. Because the vision system provides the state variables directly in the global frame, the measurement function $h(\hat{\mathbf{x}}_t)$ is the identity function:

$$\mathbf{z}_{pred} = h(\hat{\mathbf{x}}_t) = \hat{\mathbf{x}}_t \tag{9}$$

3) Linearization: Since the measurement function is the identity function, the measurement Jacobian H (the partial derivative of h with respect to the state $\mathbf{x}$) is the identity matrix:

$$H = \frac{\partial h}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{10}$$

4) Kalman Gain: The Kalman Gain K determines how much the new measurement should influence the final state estimate. It is calculated by comparing the uncertainty of the prediction $\bar{\Sigma}_t$ against the uncertainty of the measurement Q:

$$S = H \bar{\Sigma}_t H^T + Q \tag{11}$$

$$K = \bar{\Sigma}_t H^T S^{-1} \tag{12}$$

The measurement noise matrix Q represents the empirical covariance of the camera sensor system:

$$Q = \begin{bmatrix} 0.002898 & 0.000115 & 0.005556 \\ 0.000115 & 0.004000 & 0.007781 \\ 0.005556 & 0.007781 & 20.589390 \end{bmatrix} \tag{13}$$

5) State and Covariance Update: The final state estimate $\mathbf{x}_t$ is updated by applying the Kalman Gain to the innovation—the difference between the actual measurement $\mathbf{z}_t$ and the predicted measurement $\mathbf{z}_{pred}$.

$$\mathbf{x}_t = \hat{\mathbf{x}}_t + K(\mathbf{z}_t - \mathbf{z}_{pred}) \tag{14}$$

During this calculation, the angular innovation for θ is wrapped to the range $[-180, 180]$ degrees to prevent errors from angular discontinuity. Finally, the state covariance is updated to reflect the increase in certainty following the measurement:

$$\Sigma_t = (I - KH)\bar{\Sigma}_t \tag{15}$$

III. EXPERIMENT DESIGN

The experimental setup required calibrating an exteroceptive sensor and configuring the physical workspace.

A. Camera Calibration and Sensor Characterization

To ensure high-fidelity spatial tracking, the exteroceptive sensing system required rigorous calibration and environment standardization. The camera was calibrated by capturing a multitude of images of a printed checkerboard pattern, from which OpenCV was used to extract the camera intrinsics and distance coefficients, effectively rectifying lens distortion. The testing environment was constructed using a 2-meter long strip of tape bisected perpendicularly at the 1-meter mark by a 2-meter strip to provide a precise and consistent coordinate frame. To characterize sensor noise, five ArUco tags were placed at known positions on this grid to determine the absolute position and heading of the robot. By comparing these surveyed ground-truth coordinates against the poses outputted by the detection algorithm, the absolute error variances were quantified and used to define the measurement noise covariance matrix Q_t, ensuring the Extended Kalman Filter correctly weighted the reliability of the visual data during the correction step.

B. Offline Localization

The experimental evaluation comprised three distinct trajectories: a simplified linear path for baseline verification, a complex maneuver with continuous ArUco marker visibility, and a complex path designed to test filter performance during periods of partial sensor occlusion. To establish a robust benchmark for error analysis, ground truth labels were

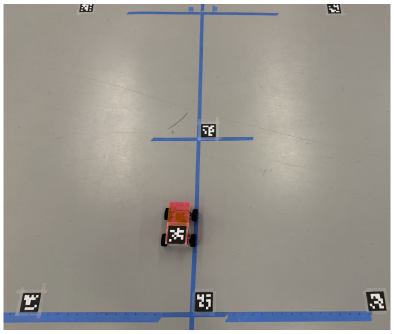

Fig. 2. Camera Capture setup for lab. 3 markers at 1m, 1 marker at 2m, and 2 markers at 3m, with the markers on the left and right being 1.8m apart

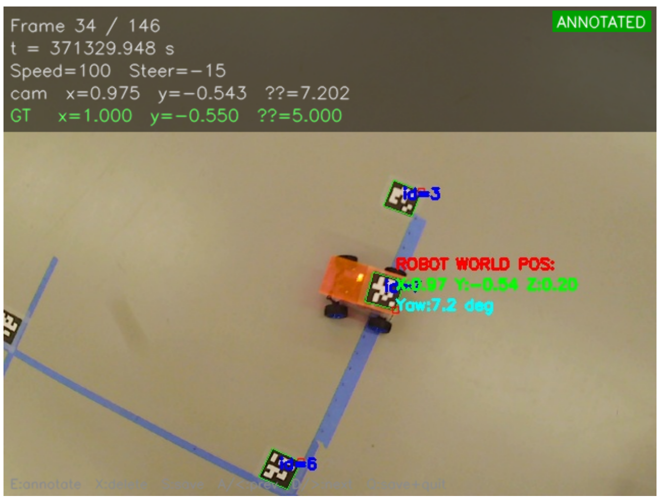

Fig. 3. Example of the Annotation GUI used to get time stamped ground truth poses for the robot.

generated by synchronizing high-resolution images with each time step and manually annotating the robot's pose relative to a coordinate system defined by physical tape markings on the laboratory floor.

C. Online Localization

The online experimental phase involved the deployment of the EKF for real-time state estimation during active robot operation. The system was configured such that the robot called the EKF update function once per control cycle. Real-time performance was visualized via a GUI, displaying the live state estimate and its associated confidence ellipse on the camera feed. To document the system limits, a trajectory was driven that forced the robot in and out of the camera's field of view. The absolute position was determined by taking an image at every frame, then using the grid to determine the real physical position of the robot. This and the data from the filter were combined to provide a qualitative and quantitative comparison between the physical robot movement and the filter's perceived state, demonstrating the system's ability to maintain localization under dynamic and partially observable conditions.

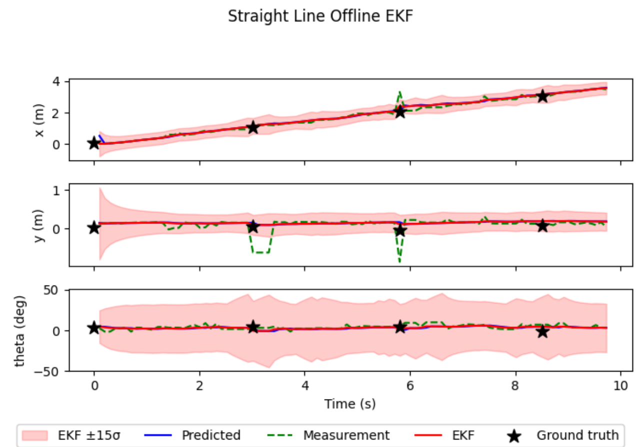

Fig. 4. EKF output for following a straight trajectory with (mainly) complete observation using the Aruco markers. Discontinuities in the Aruco detection occurred where the robot passed over the reference markers.

IV. EXPERIMENTAL RESULTS

In the following subsections the results of the experiments are presented.

A. Sensor Calibration

The camera calibration process yielded a camera matrix with focal lengths f_x≈1538.79 and f_y≈1537.26, and principal point coordinates at (829.21,577.15). The radial distortion was primarily characterized by a k_1 coefficient of -0.3965 and k_2 of 0.1470. Despite this calibration, it was observed that the camera was poorly able to resolve ArUco tags at the periphery of the frame. This spatial inconsistency necessitated a custom variance setup where measurement noise was profiled at five distinct grid locations. By using these empirical variances to populate Q_t rather than relying on a uniform assumption, the EKF maintained stability even as detection quality degraded near the frame boundaries.

B. Sensor Noise and Outlier Mitigation

During dynamic testing, the raw ArUco detection data exhibited high-frequency noise and intermittent outliers. These "jumps" in the measurement vector z_t were frequently as large as ±10cm, which would have caused significant instability in the state estimate if left unaddressed. To mitigate this, a median outlier filter was implemented to pre-process the camera data. This filter effectively rejected readings that deviated from the median of a 10 frame sliding window. The inclusion of this filter significantly smoothed the measurement input, resulting in a stable state estimate even in suboptimal lighting conditions.

C. Offline Localization

The EKF was evaluated using data logged from a differential drive robot tracked by an overhead camera and ArUco markers. The analysis compared the EKF state estimates against ground truth across three trajectories: a simple straight line, a curved path, and a complex multi-directional run.

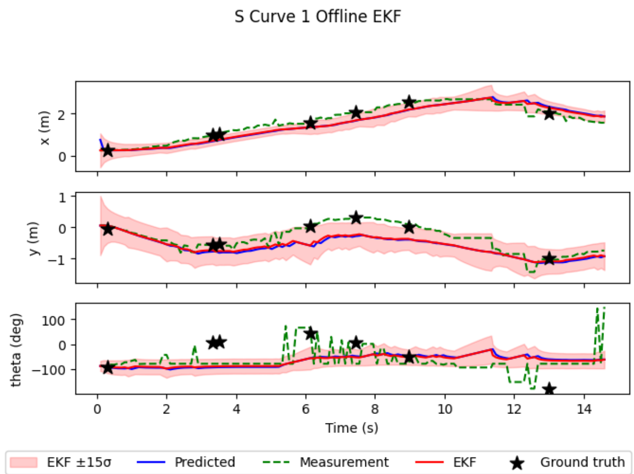

Fig. 5. Complex curve with high observation of the Aruco makers. Variance in the angle estimation can be observed from the camera based detection.

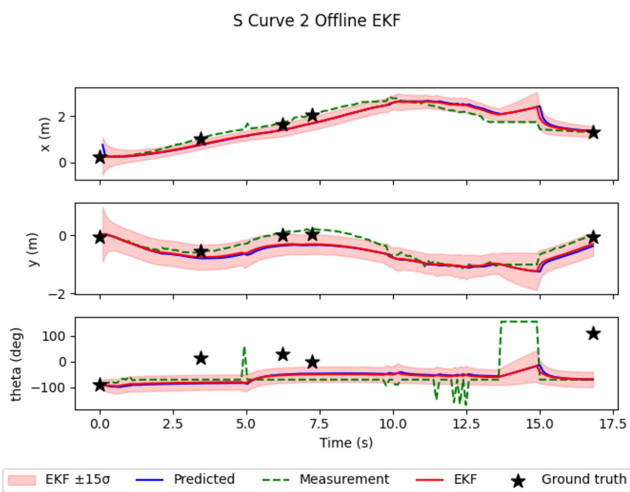

Fig. 6. Complex curve with a planned period of non-observation using the camera from 13-15 seconds. Here it can be observed that the confidence decreases as the correction step is omitted.

1) Accuracy and Error Summary: Across all experimental trials, the EKF maintained a high level of positional accuracy. In straight-line tests, the filter achieved a mean error of 0.1061 ± 0.1134 m in x and 0.1021 ± 0.0282 m in y, with a very stable heading error of $3.1336 \pm 1.1629°$. As the complexity of the maneuvers increased in the curved and multi-directional runs, the lateral and longitudinal errors grew slightly, ranging between 0.12 m and 0.24 m.The most significant deviation occurred in the orientation (θ) during high-curvature turns, where mean errors reached $86.0398 \pm 55.4595°$ and $80.8873 \pm 53.6572°$. This suggests that while the kinematic bicycle model effectively tracks translation, the rotational component is highly sensitive to the rapid changes in steering angle α and potential marker occlusion during sharp rotations.

2) Filter Behavior and Robustness: When the correction step was disabled, the filter relied solely on dead reckoning. This resulted in a continuous growth of the state covariance Σ_t, visualized as expanding confidence ellipses. Upon enabling the correction step, the ArUco measurements successfully ”collapsed” these ellipses, resetting the robot’s pose and preventing the accumulation of integration drift.In-Frame vs. Out-of-Frame: The filter demonstrated robust behavior when the robot moved out of the camera’s field of view. During these intervals, the EKF transitioned to a pure prediction mode, propagating the state using only encoder counts and steering commands. The covariance matrix increased during this time, reflecting growing uncertainty. Once the robot returned to the frame, the first available ArUco measurement provided a correction that instantly reduced the uncertainty and realigned the state estimate with the truth.Initialization: The EKF proved robust to ”unknown” start conditions. By initializing the filter with a large initial covariance Σ_0 and an incorrect pose, the Kalman Gain K placed high weight on the initial camera measurements. This allowed the filter to rapidly converge from a bad initial guess to the correct global pose within the first few frames of data.

D. Online Localization

Experimental validation of the online localization system demonstrated high spatial fidelity, particularly when the robot remained within the exteroceptive sensor’s primary field of view. Under these conditions, the EKF maintained a maximum lateral deviation of 5.0 cm relative to the ground-truth ArUco markers. Quantitative analysis of the trajectory yielded a Root Mean Square Error (RMSE) of approximately 1.31 cm (MSE=1.7058). This low error magnitude confirms the filter’s efficacy in suppressing measurement noise while maintaining an accurate global pose estimate.

E. Uncertainty and Correlation Analysis

Using 4162 samples of the robot in 5 different poses, the covariance of the measurement vector z_t was calculated to be:

$$\Sigma = \begin{bmatrix} 0.002898 & 0.000115 & 0.005556 \\ 0.000115 & 0.004000 & 0.007781 \\ 0.005556 & 0.007781 & 20.589390 \end{bmatrix} \tag{1}$$

The diagonal elements revealed high positional confidence (x, y), but the yaw variance was significantly higher at 20.589. This identifies heading estimation as the primary source of uncertainty, likely due to pixel fluctuations in tag corner detection. A stronger correlation was noted between yaw and the y-coordinate (0.007781) compared to the x-coordinate (0.005556), consistent with the non-linear coupling of the bicycle model during lateral maneuvers.

V. CONCLUSION

The implementation of the Extended Kalman Filter (EKF) successfully demonstrated a robust framework for real-time localization of a car-like mobile robot by mathematically fusing a non-linear bicycle motion model with exteroceptive visual measurements. This integration effectively mitigated the compounding errors inherent in pure dead reckoning, providing a high degree of precision within the camera’s field of view and significantly outperforming the open-loop

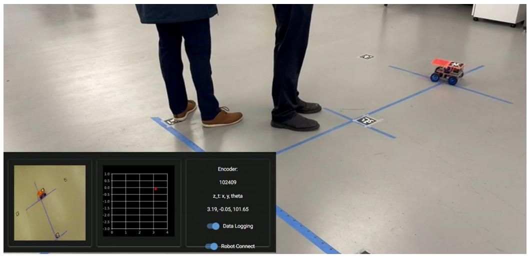

Fig. 7. Location of robot in GUI overlayed onto physical location. Robot closely tracks linearization plot, around 2cm visible error

odometry models developed in previous work. The EKF further demonstrated its reliability by successfully recovering from large deliberate initial pose errors and maintaining stable uncertainty bounds during intervals of total sensor occlusion, confirming that the filter is a vital mechanism for maintaining state estimation integrity in partially observable environments.

VI. FUTURE WORK

Our main goal is to replace the steering assembly to approximate an Ideal Ackerman drive [3]. This would allow us to drastically reduce front wheel slippage, thus making our odometry more reliable. In addition, further adjustments need to be made to the relative output velocities between the back two wheels in order to remove the bias in the current steering implementation. Finally we hope to build a suite of more complete analysis and testing tools, to allow for further profiling of the specific behaviors of the robot.

VII. REFERENCES

REFERENCES

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